Steam Power Plants

Combustion

Combustion is the burning of fuels; typically hydrocarbons, with oxygen to release large quantities of heat. Fuels may also contain other elements like sulfur and other substances, called incombustibles. These fuels can be in either solid, liquid or gaseous states, with gaseous fuels like LPG or naphtha containing minimal sulfur. For oil fuels, sulfur content ranges from 0.5% to 4%.

Fuels can be quantified by their caloric value, which is the amount of heat energy produced when 1 unit (kg or m^3) of fuel is completely burnt (depends on state), in units kJ/kg or kJ/m^3. For fuels containing hydrogen, there are two caloric values depending if water is produced in a liquid or vapour state. If the fuel is burnt completely and the products are then cooled on 25C to allow for water to condense, the amount of heat released is the gross or higher caloric value (HCV). Otherwise, if the products are not let to be cooled, the heat released is the net or lower caloric value (LCV). Since more heat is produced during the cooling phase $HCV > LCV$. However, it is usually impractical for the fuel to cool to reach the HCV, and the LCV is a more practical result. The HCV and LCV of fuels are related as so

$$HCV = LCV + m \times h_{fg}$$

where h_{fg} is the specific latent heat of evaporation of water.

These fuels can be in either solid, liquid or gaseous states, and hence come from different sources. Solid fuels typically form from plant material which has absorbed energy from the sun, like wood and coal. Their HCV value ranges from 16 to 35 MJ/kg. Liquid fuels come from crude petrolum or plant matter, like petrol, diesel, oil and alcohol. These fuels typically have HCVs of 44 to 46.8 MJ/kg. Gaseous fuels either occur naturally like from natural gases or are manufactured from coal and petroleum, like propane. Their HCVs range much more, from 6 to 120MJ/m^3.

One additional note about liquid fuels is that they can "burn" at different points. With a spark or flame, liquid fuels need to reach the flash point to be able to ignite, which is important to consider when transporting fuels. Liquid fuels can also spontaneously ignite without needing a spark or flame at high enough temperatures, minimally needing the ignition temperature.

To perform analysis on fuels, stoichiometric analysis is required, by finding the amount of products formed when fully combusted or not. In general, the principle is that each element reacts a certain way during combustion. In complete combustion,

$$2H_2 + O_2 \rightarrow 2H_2O$$
$$C + O_2 \rightarrow CO_2$$
$$S + O_2 \rightarrow SO_2$$
$$2CO + O_2 \rightarrow 2CO_2$$

but if there is insufficient oxygen or improper mixing of fuel, incomplete combustion occurs, causing carbon to combust differently

$$2C + O_2 \rightarrow 2CO$$

while also forming excess carbon.

Using these equations, the combustion equation can be formed. In general, for a fuel containing carbon, hydrogen, oxygen, nitrogen and sulfur,

$$aC + bH_2 + cO_2 + dS + eN_2 + y(0.21O_2 + 0.79N_2) \\ \rightarrow \alpha CO_2 + \beta H_2O + \gamma SO_2 + (0.79y + e)N_2$$

and by balancing the equation, we get that

$$\alpha = a, \beta = b, \gamma = d$$

and that the amount of air used, y, can be found from

$$c + 0.21y = \alpha + \frac{1}{2}\beta + \gamma$$

By adding terms at the products side, extra oxygen or carbon can be solved for in the equation.

The particular value of y found from the earlier equation is also called the stoichiometric air, and when converted into kg, can be divided by the mass of fuel used to get the stoichiometric air to fuel ratio. This value represents the minimum amount of air needed for complete combustion.

Since most practical cases do not supply the exact amount of air needed for complete combustion with no excess, the $\%\text{Excess Air}$ can be used to determine how much excess air was used.

$$\begin{aligned}\%\text{ excess air} &= \frac{\text{Actual Air} - \text{Stoichiometric Air}}{\text{Stoichiometric Air}} \\ &= \frac{\text{Actual A/F} - \text{Stoichiometric A/F}}{\text{Stoichiometric A/F}}\end{aligned}$$

Excess air combustion is usually called “weak” or “lean” mixtures. If the number is negative, the fuel undergoes incomplete combustion and the mixture is called “rich”.

Another useful formula is Dulong’s formula, which estimates the HCV of solid and liquid fuels.

$$HCV = 4.19\left(80.8C_{\%} + 345\left(\frac{H_{\%} - O_{\%}}{8} + 22.2S_{\%}\right)\right) kJ/kg$$

where the formula only accepts percent mass. This can then be used as a approximation of LCV too.

Steam Distribution & Nozzles

Steam distribution is critical in steam power plants to ensure efficiency between different equipment. Steam distribution is usually controlled by either pipe sizing, or mains drainage (to other components).

Steam pipes need to be correctly sized to transport the correct amount of steam, while also reducing heat loss due to radiation for too large of diameters. Hence, pipe sizing is determined either by pressure or by velocity.

If the steam pressure required is lower than the boiler pressure, and the pipe is long, sizing has to be done. This requires knowing the pressures and flowrates at both equipment, and the length of the pipe connecting the two.

Firstly, a lookup table is needed to convert pressures into a pressure factor.

TABLE 1 Pressure Factors for Pipe Sizing

Pressure bar abs	Volume m^3/kg	Pressure Factor	Pressure bar gauge	Volume m^3/kg	Pressure Factor	Pressure bar gauge	Volume m^3/kg	Pressure Factor
0,05	28,192	0,0301	2,15	0,576	9,309	7,70	0,222	66,31
0,10	14,674	0,0115	2,20	0,568	9,597	7,80	0,219	67,79
0,15	10,022	0,0253	2,25	0,660	9,888	7,90	0,217	69,29
0,20	7,649	0,0442	2,30	0,552	10,18	8,00	0,215	70,80
0,25	6,204	0,0681	2,35	0,544	10,48	8,10	0,212	72,33
0,30	5,229	0,0970	2,40	0,536	10,79	8,20	0,210	73,88
0,35	4,530	0,1308	2,45	0,529	11,40	8,30	0,208	75,44
0,40	3,993	0,1694	2,50	0,522	11,41	8,40	0,206	77,02
0,45	3,580	0,2128	2,55	0,515	11,72	8,50	0,204	78,61
0,50	3,240	0,2610	2,60	0,509	12,05	8,60	0,202	80,22
0,55	2,964	0,3140	2,65	0,502	12,37	8,70	0,200	81,84
0,60	2,732	0,3716	2,70	0,496	12,70	8,80	0,198	83,49
0,65	2,535	0,4340	2,75	0,489	13,03	8,90	0,196	85,14
0,70	2,365	0,5010	2,80	0,483	13,37	9,00	0,194	86,81
0,75	2,217	0,5727	2,85	0,477	13,71	9,10	0,192	88,50
0,80	2,087	0,6489	2,90	0,471	14,06	9,20	0,191	90,20
0,85	1,972	0,7298	2,95	0,466	14,41	9,30	0,189	91,92
0,90	1,869	0,8153	3,00	0,461	14,76	9,40	0,187	93,66
0,95	1,777	0,9053	3,10	0,451	15,48	9,50	0,185	95,41
1,013	1,673	1,025	3,20	0,440	16,22	9,60	0,184	97,18
bar gauge			3,30	0,431	16,98	9,70	0,182	98,96
0	1,673	1,025	3,40	0,422	17,75	9,80	0,181	100,75
0,05	1,601	1,126	3,50	0,413	18,54	9,90	0,179	102,57
0,10	1,533	1,230	3,60	0,405	19,34	10,00	0,177	104,40
0,15	1,471	1,339	3,70	0,396	20,16	10,20	0,174	108,10
0,20	1,414	1,453	3,80	0,389	21,00	10,40	0,172	111,87
0,25	1,361	1,572	3,90	0,381	21,85	10,60	0,169	115,70

Then, an expression is used to find a pressure drop factor, given as

$$F = \frac{P_1 - P_2}{L_{\text{total}}}$$

where P1 and P2 are the pressure factors and L total is the total length of travel, with some allowance. More specifically, L total is

$$L_{\text{total}} = \begin{cases} L_0 + 0.1L_0 \text{ when } L_0 > 100m \\ L_0 + 0.2L_0 \text{ when } L_0 < 100m \end{cases}$$

After finding the pressure drop factor F, another lookup table is used to convert it into the diameter and flow rate of the pipe.

TABLE 2 Pipeline Capacity and Pressure Drop Factors

Factor F		Pipe Size in mm 15	20	25	32	40	50	65	80	100	125	150	175	200	225	250	300
0.00016	x						30.40	55.41	90.72	199.1	360.4	598.2	890.0	1275	1755	2329	3800
	y						4.30	4.86	5.55	6.82	7.90	9.16	10.05	10.94	11.94	12.77	14.54
0.00020	x					16.18	34.32	62.77	103.0	225.6	407.0	662.0	1005	1437	1966	2623	4276
	y					3.96	4.85	5.51	6.31	7.72	8.92	10.13	11.34	12.33	13.37	14.38	16.36
0.00025	x				10.84	17.92	38.19	69.31	113.2	249.9	450.3	735.5	1108	1678	2183	2904	4715
	y				3.74	4.39	5.40	6.08	6.92	8.56	9.87	11.26	12.51	14.40	14.85	15.92	18.04
0.00030	x				11.95	19.31	41.83	75.85	124.1	271.2	491.9	804.5	1209	1733	2390	4172	5149
	y				4.13	4.73	5.92	6.65	7.60	9.29	10.79	12.31	13.65	14.87	16.26	17.39	19.07
0.00035	x			6.86	12.44	20.59	43.76	80.24	130.01	285.3	519.2	845.3	1279	1823	2497	3346	5406
	y			3.88	4.30	5.04	6.21	7.04	7.96	9.77	11.38	12.94	14.44	15.64	17.00	18.34	20.69
0.00045	x		3.62	7.94	14.56	23.39	50.75	92.68	150.9	333.2	604.6	979.7	1478	2118	2913	3884	6267
	y		3.54	4.49	5.03	5.73	7.18	8.13	9.24	11.42	13.26	15.00	16.68	18.18	19.82	21.29	23.99
0.00055	x		4.04	8.99	16.18	26.52	57.09	103.8	170.8	373.1	674.2	1101	1663	2382	3281	4338	7057
	y		3.96	5.09	5.59	6.49	8.08	9.10	10.46	12.78	14.78	16.85	18.77	20.44	22.32	23.78	27.01
0.00065	x		4.46	9.56	17.76	29.14	62.38	113.8	186.7	409.8	739.9	1207	1823	2595	3597	4781	7741
	y		4.37	5.41	6.13	7.14	8.82	9.98	11.43	14.04	16.22	18.48	20.58	22.27	24.47	26.21	29.62
0.00075	x		4.87	10.57	19.31	31.72	68.04	124.1	203.2	445.9	804.5	1315	1977	2836	3908	5172	8367
	y		4.77	5.98	6.67	7.77	9.62	10.88	12.44	15.28	17.64	20.13	22.32	24.34	26.59	28.35	32.02
0.00085	x		5.52	11.98	21.88	35.95	77.11	140.7	230.2	505.4	911.8	1490	2240	3215	4429	5861	9482
	y		5.41	6.78	7.56	8.80	10.91	12.34	14.09	17.32	19.99	22.81	25.29	27.59	30.13	32.13	36.29
0.00100	x	1.96	5.84	12.75	23.50	38.25	81.89	148.6	245.2	539.4	968.5	1579	2403	3383	4707	6228	10052
	y	4.10	5.72	7.21	8.12	9.37	11.58	13.03	15.01	18.48	21.24	24.17	27.13	29.03	32.02	34.14	38.47
0.00125	x	2.10	6.26	13.57	24.96	40.72	87.57	159.8	261.8	577.9	1038	1699	2544	3634	5035	6655	10639
	y	4.39	6.13	7.68	8.62	9.97	12.39	14.02	16.03	19.80	22.76	26.01	28.72	31.19	34.26	36.48	40.71
0.00150	x	2.39	7.35	15.17	28.04	45.97	98.84	179.3	295.1	652.8	1172	1908	2896	4091	5631	7493	11999
	y	5.00	7.20	8.58	9.68	11.26	13.98	15.72	18.07	22.37	25.70	29.21	32.69	35.11	38.31	41.08	45.92
0.00175	x	2.48	7.51	16.30	29.61	49.34	103.4	188.8	311.1	686.5	1270	2017	3046	4291	5921	7852	13087
	y	5.19	7.36	9.22	10.23	12.08	14.63	16.56	19.05	23.52	27.85	30.88	34.39	36.83	40.28	43.04	50.08
0.0020	x	2.84	8.58	18.63	33.83	56.39	118.2	215.8	355.5	784.6	1451	2305	3482	4904	6767	8974	14956
	y	5.94	8.40	10.54	11.68	13.81	16.72	18.93	21.77	26.88	31.82	35.28	39.31	42.09	46.04	49.19	57.24
0.0025	x	3.16	9.48	20.75	37.25	61.30	132.0	240.5	391.3	881.7	1556	2456	3819	5422	7544	10090	16503
	y	6.61	9.29	11.74	12.86	15.01	18.67	21.09	23.96	30.21	34.12	38.97	43.11	46.53	51.33	55.31	63.16
0.0030	x	3.44	10.34	22.5	40.45	66.66	143.4	262.0	429.8	924.4	1701	2767	4183	6068	8275	11033	18021
	y	7.20	10.13	12.73	13.97	16.33	20.29	22.98	26.32	32.29	37.30	42.36	47.22	52.08	56.30	60.48	68.97

The two values in each cell represent different values, the x value is the mass flowrate in kg/h, while the y value is the m/s velocity for a 1m^3/kg volume of steam. Hence, the correct value needs to be picked depending on the givens.

Hence, from the factor F, the smaller or larger closer value is picked depending on if there is a pressure drop or increase respectively. Then, by picking the right flowrate based on the givens, the suitable diameter can be picked.

In some scenarios, sizing by pressure is not practical as the diameter found would cause extreme velocities, which can cause erosion, due to extreme pressure differences. Thus, sizing by picking a reasonable velocity may be done instead. A reasonable velocity for saturated steam is usually about 15 to 40m/s. Above these velocities, erosion is likely to occur, especially as x decreases.

Sizing by velocity is also much easier to find, by knowing the velocity, pressure, and flow rate, the suitable diameter can be read off from the table.

TABLE 3 Pipe Line Capacities at Specific Velocities

Pressure bar	Velocity m/s	15mm	20mm	25mm	32mm	40mm	kg/h 50mm	65mm	80mm	100mm	125mm	150mm
0.4	15	7	14	24	37	52	99	145	213	394	648	917
	25	10	25	40	62	92	162	265	384	675	972	1457
	40	17	35	64	102	142	265	403	576	1037	1670	2303
0.7	15	7	16	25	40	59	109	166	250	431	680	1006
	25	12	25	45	72	100	182	287	430	716	1145	1575
	40	18	37	68	106	167	298	428	630	1108	1712	2417
1.0	15	8	17	29	43	65	112	182	260	470	694	1020
	25	12	26	48	72	100	193	300	445	730	1160	1660
	40	19	39	71	112	172	311	465	640	1150	1800	2500
2.0	15	12	25	45	70	100	182	280	410	715	1125	1580
	25	19	43	70	112	162	295	428	656	1215	1755	2520
	40	30	64	115	178	275	475	745	1010	1895	2925	4175

As said earlier, steam distribution is also controlled by mains drainage. More specifically, condensed water from steam has to be drained out as steam inevitably loses heat.

One way to control this is by preventign the buildup of water by ensuring the mains do not flow downward by a long distance, more specifically that is does not have a fall of more than 12mm in 3m, in the direction of steam flow. If this is the case, condensate collects at the bottom and makes it harder to remove from the pipe. This also encourages waterhammer and mixing of the water with steam, causing very wet steam.

However, this can be used to our advantage, by letting the direction fall to collect condensate, and placing drain points are those locations, usually at intervals 30m to 50m along the length of the main, at low points.

For drain pipes, the diameter needs to have equal diameter to the main pipes, or at least up to 100mm bores. Above 100mm bores, the drain pipe’s bore may be smaller, like using a 100mm pocket for a 150mm line, 150mm for 200mm, etc.

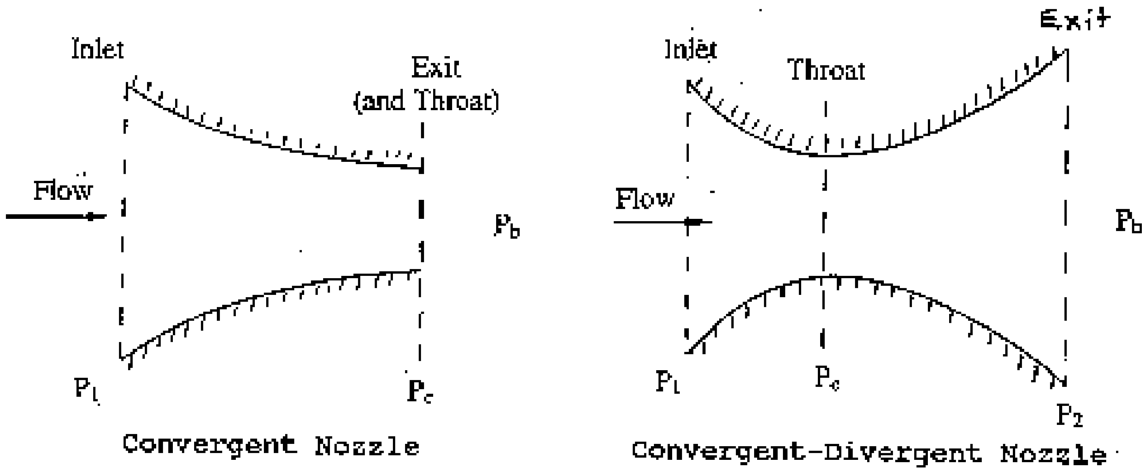
These pockets may also have steam traps, automatic valves that open when condensate flows but not when steam is in the drain pipe, to prevent steam from escaping.

For water particles in the air carried by steam flow, they do not collect and need to be removed by steam driers or separators. These are typically placed right before equipment that needs to use steam, or steam reducing valves, to ensure it can remove as much moisture as possible. These work on the principle that water has a higher density than steam, by flinging the heavier water particles out of the system as they impact on baffle plates and on the sides of the vessel.

Similar to why steam driers are placed at the end of steam pipes, steam branches should be taken from the top of the supply, to ensure the driest steam is sourced.

One common steam apparatus are steam nozzles, which increase the velocity of steam for apparatus like turbines. Generally, they work on the principle of converting pressure and temperature into speed, via the geometry of the nozzle.

Generally, there are two types of nozzles, convergent and convergent-divergent nozzles.



Both nozzles have some inlet pressure that steam enters from, and they reach a choke point, where the pressure reaches a critical point. From here, C-D nozzles continue to drop in pressure until it reaches the outlet pressure. In some cases, there may be some back pressure, but in most cases the back pressure is equal to the outlet pressure.

We already know that when cross sectional area drops, velocity increases, and pressure decreases, hence why the convergent nozzle increases velocity. However, this is only applicable for subsonic flow, when the steam is moving slower than the speed of sound.

However, when the steam reaches fast enough speeds to reach the speed of sound, which we say that the flow is "choked", further decrements in area does not speed up the steam. Instead, to reach supersonic speeds, the steam must be allowed to expand to increase the velocity, hence why convergent-divergent nozzles are used so often.

We often refer to subsonic and supersonic flow by the mach number. When the flow is slower than the speed of sound, $M < 1$. When the flow is choked, $M = 1$, and when the flow is supersonic, $M > 1$.

Type of Nozzle \ Flow Speed	**Sub-sonic**	**Supersonic**
Convergent	Velocity increases (Acceleration) Pressure decreases (Expansion)	Velocity decreases (Deceleration) Pressure increases (Compression)
Divergent	Velocity decreases (Deceleration) Pressure increases (Compression)	Velocity increases (Acceleration) Pressure decreases (Expansion)

At the choke of C-D nozzles, the steam reaches its critical point, when flowrate is at its maximum. Because of steam flowing quickly through the nozzle at the start, we assume it to flow isentropically. Thus, if isentropic flow follows expansion of $pv^k = \text{const}$, where k is a steam index, the ratio of the pressure at the throat against the inlet pressure is

$$\frac{P_c}{P_1} = \left(\frac{2}{k+1}\right)^{\frac{k}{k-1}}$$

where k is 1.3 for superheated steam, and 1.135 for wet steam.

We can also find the mass flow rate of steam through the nozzle. The theoretical mass flow rate through the nozzle is

$$\dot{m} = \frac{CA}{v} = \frac{\text{velocity} \times \text{cross sectional area}}{\text{specific volume}}$$

For the specific volume, it is either determined through steam tables for wet steam, via

$$v_{\text{wet steam}} = xv_g$$

or by using an empirical formula for superheated steam, as

$$v_{\text{superheated}} = \frac{2.308(h[kJ/kg] - 1943)}{P[bar] \times 10^3}$$

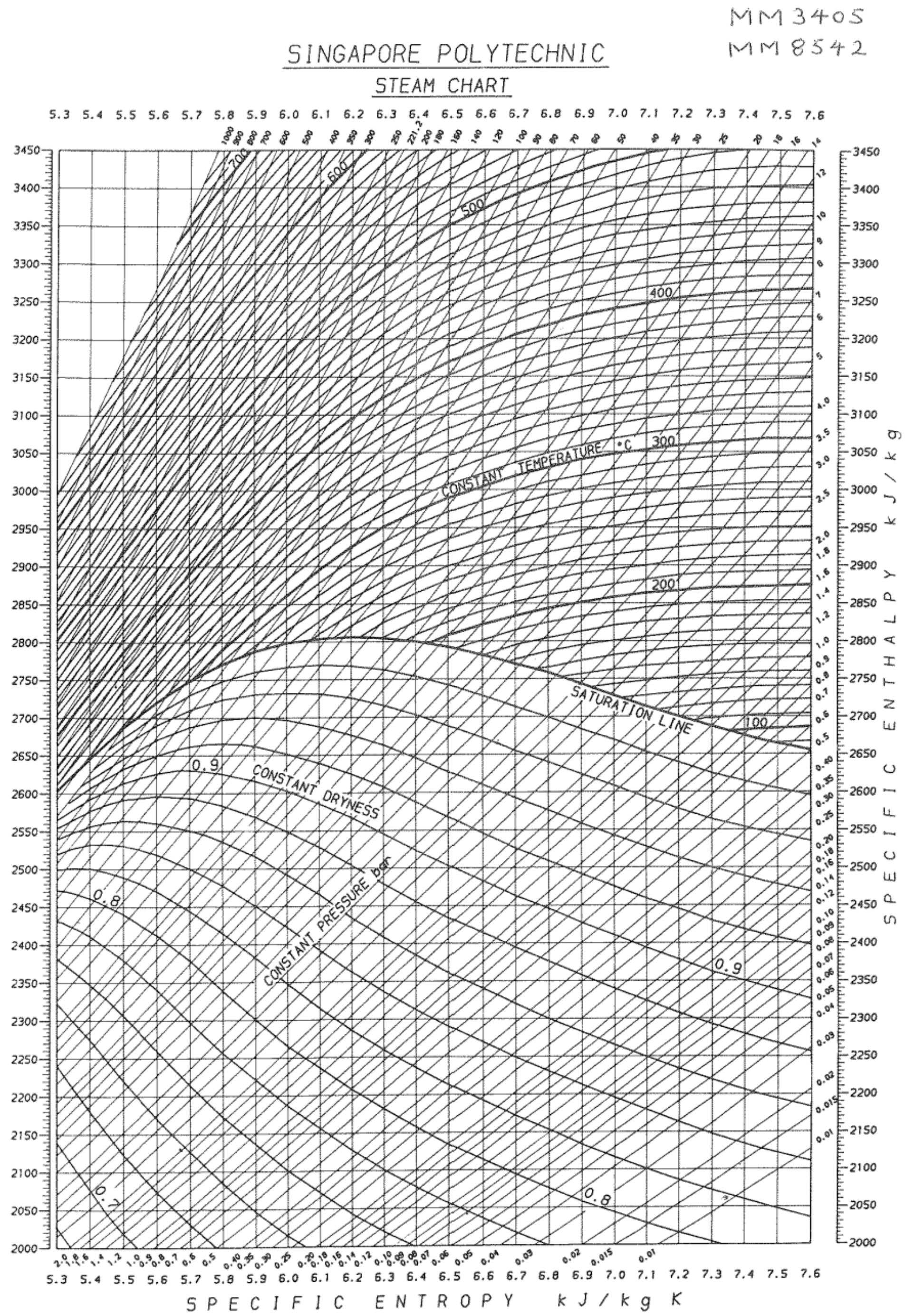
The velocity of the steam is also determined through

$$c = \sqrt{2(h_1 - h)}$$

where h1 and h are in J/kg, and the initial speed is assumed to be 0. For kJ/kg, the formula is

$$c = \sqrt{2000(h_1 - h)}$$

where h is the enthalpy at a picked point. These enthalpies are read from a steam chart, which is a h-s steam chart that also has lines representing constant pressure and temperature lines.



In ideal cases, we assume that flow through the whole nozzle is isentropic, however, because $\Delta S > 0$, the process does not follow a vertical line on the graph, but instead veers to the right a bit. Hence, the mass flow rate differs slightly, by

$$C_d = \frac{\dot{m}_{\text{actual}}}{\dot{m}_{\text{isen}}}$$

where C_d is known as the coefficient of discharge. The velocity differs slightly at the outlet too, since the enthalpy is slightly higher than the ideal. Thus,

$$C_v = \frac{C_2'}{C_2}$$

where C_v is the coefficient of velocity.

C_2' is also determined from $\sqrt{2000(h_1 - h_2')}$

Generally, for steam nozzle problems, the pressures of the inlet and outlet are given, as well as the inlet temperature. Knowing the inlet temperature and pressure, the inlet enthalpy can be found, and hence the process line by drawing a vertical line on the steam chart. This allows for values like velocity, specific volume, and hence mass flow rate to be found, depending on the question.

When considering practical values like actual velocity C_2', the enthalpy can be found from the new velocity, and by knowing the end pressure, the corresponding point on the steam chart can be found.

In some nozzles, there may be a certain back pressure applied at the end, to satisfy certain design conditions, being,

- the flow is choked, i.e. $M = 1$ at the throat.
- the nozzle is able to expand the fluid throughout the whole length of the nozzle.
- the pressure at exit is equal to the back pressure

Both the last two conditions help to ensure shockwaves do not happen as the flow reaches supersonic speeds.

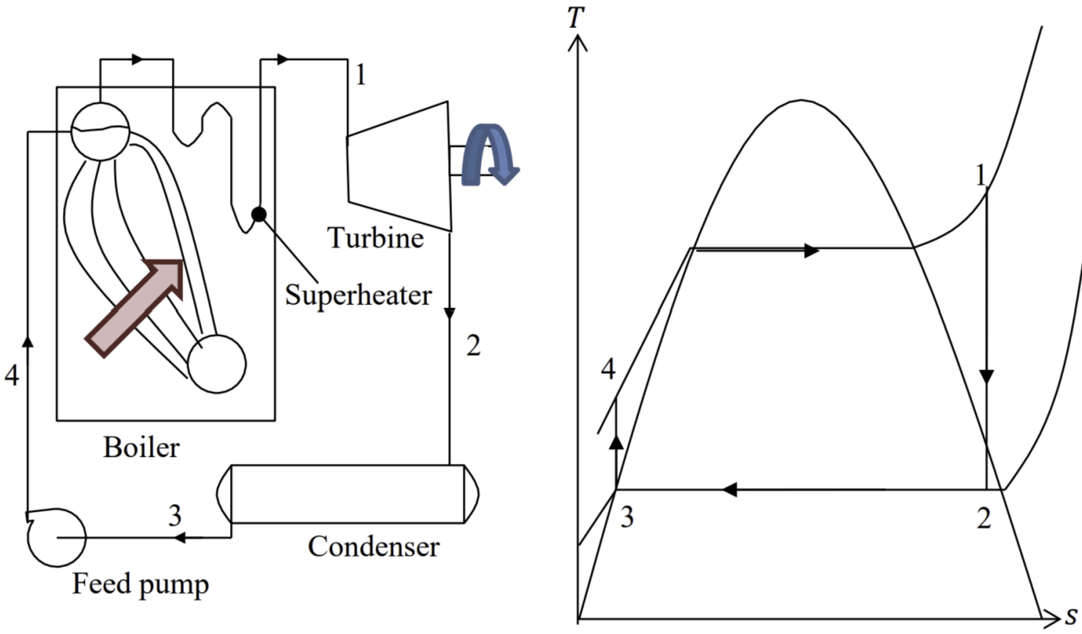
If the nozzle is poorly designed, the steam may undergo over or underexpansion.

Overexpansion is when the back pressure is higher than the designed back pressure, causing mass flow rate and exit velocity to be below the optimal values.

Underexpansion is when the back pressure is below the designed back pressure, causing damaging shockwaves to occur outside the nozzle as steam exits the nozzle.

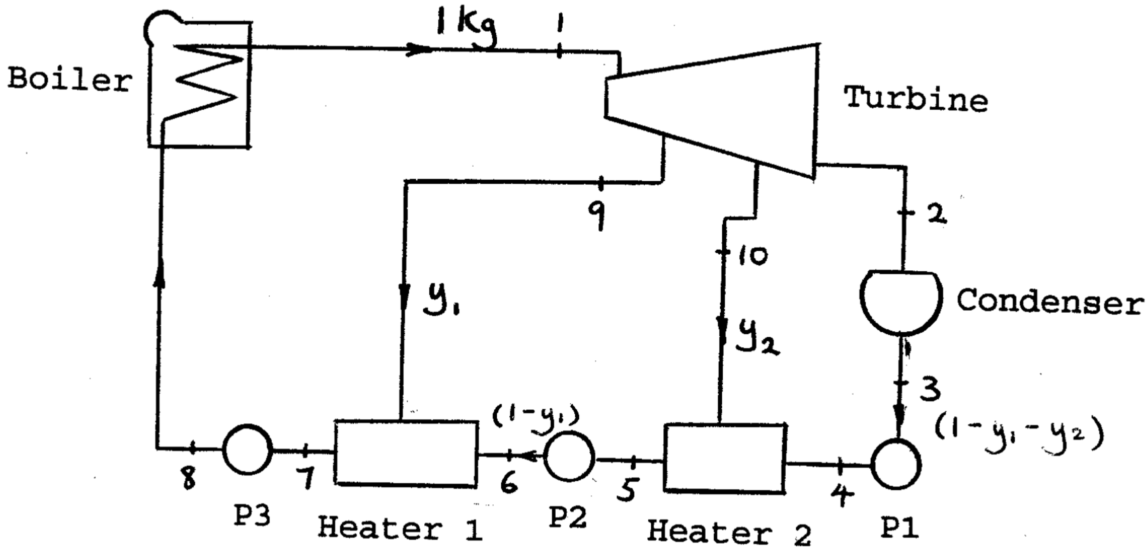
Regenerative Cycle

Recall the standard Rankine cycle from Thermofluids II.



One major shortcoming of this is that all the condensed steam flows into the boiler, meaning the boiler must supply a high amount of energy to heat and boil the condensed steam. Thus, one way to mitigate this is by allowing some steam to bleed off from the turbine, to then be heated and passed back into the boiler, which reduces the amount of heat needed to boil the condensed steam back. This is the main concept of regenerative cycles, which allow some steam to bleed off from the turbine at different stages into separate heaters and pumps to be fed back into the turbine.

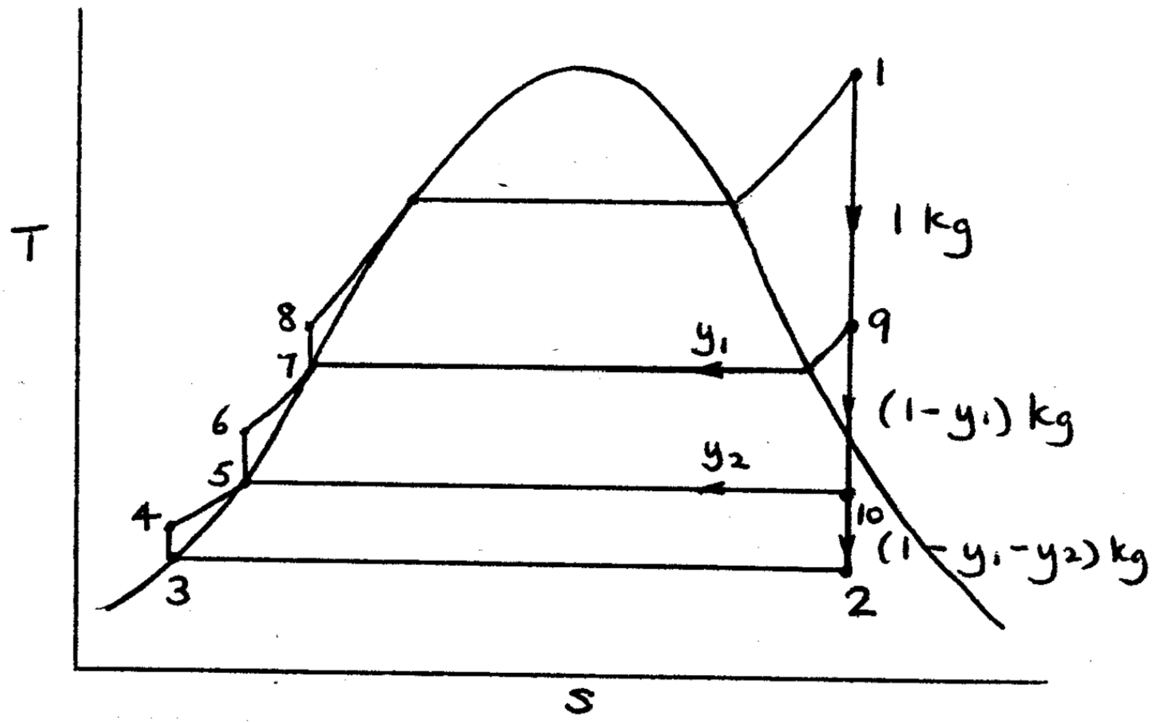
A typical regenerative steam power plant has the following setup:



After the steam has boiled, it expands isentropically in the turbine, and some steam is bled off at a lower pressure into a heater. As the steam continues to expand, more steam bleeds off into other heaters, until the remaining steam exits the turbine and reaches the condenser, where pumps push the cooler steam through the heaters and back into the boiler.

Consider the output steam from the boiler, which has a certain pressure P_1 and temperature T_1, hence having an enthalpy of h_1. As it expands through the turbine, some steam bleeds off at a pressure P_9 and P_{10}, until the rest of the steam reaches the lowest pressure of P_2. Hence, $P_1 > P_9 > P_{10} > P_2$. From here, the typical Rankine cycle continues from stage 2, where steam goes into the condenser, before a pump pumps it back into the boiler. However, the bled off steam also flows into the boiler into a different manner. Consider the steam at stage 9. Since it is bled off from the earliest portion of the turbine, it is the hottest, and heat is transferred in heater 1, heating the incoming steam from other portions of the cycle before being pumped into the boiler. This similar process happens for stage 10, where it helps to heat the incoming steam slightly before being pumped back to the boiler.

On the T-S diagram, the different cycles are shown.



Notice how this diagram makes some assumptions, firstly that the turbine expansion is perfectly isentropic, and that the heater outlets always produce saturated water at the same pressure of the inlets. This also shows that the heaters need to have the same inlet pressures too. Another assumption made in calculations is that the pump work is negligible, and can be ignored for metrics like efficiency.

Although we may know the pressure that the steam bleeds off at each section, the proportion of steam bled off is not directly known. Instead, we can use conservation of energy and mass to get the proportion of steam in each section. By assuming 1kg unit of steam is in the system, we can assume that a proportion of steam, y_1, is bled off from the turbine, followed by another proportion of steam, y_2, is bled off. Hence, the remaining steam exiting the turbine is $1 - y_1 - y_2$.

From here, the mass of steam flowing through can be determined easily, since components with one inlet and outlet must pass the same amount of steam through it. As for the heaters, the mass flowing in is different. Consider heater 2, with y_2 amount of steam bled off from the turbine and $(1 - y_1 - y_2)$ amount of steam from the exit of the turbine and condenser, both flowing into the heater. By conservation of mass, the exit amount will be

$$y_2 + (1 - y_1 - y_2) = 1 - y_1$$

and likewise for heater 1, since $1 - y_1$ and y_1 amount of steam are fed into the heater. However, to find the actual numerical values of y_1 and y_2, we consider the energy conservation in each heater. Since the energy in must equal the energy out, we consider the net enthalpy in and out of the heaters. We already know that the enthalpy of the boiler exit h_1 can be determined, and since the turbine expansion is isentropic, the enthalpies of the steam exiting the turbine, being h_9, h_{10} and h_2, can be found too. Since we are also assuming that h_3, h_5 and h_7, the steam exiting the condenser and heaters respectively are saturated, they can be found too, and the enthalpies after the pumps h_4, h_6 and h_8 can be assumed to be equal to the enthalpy of the pump inlet due to pump work being neglected, every enthalpy can be found.

Thus, considering the net enthalpy going in and out of heater 1 for example, it is given as

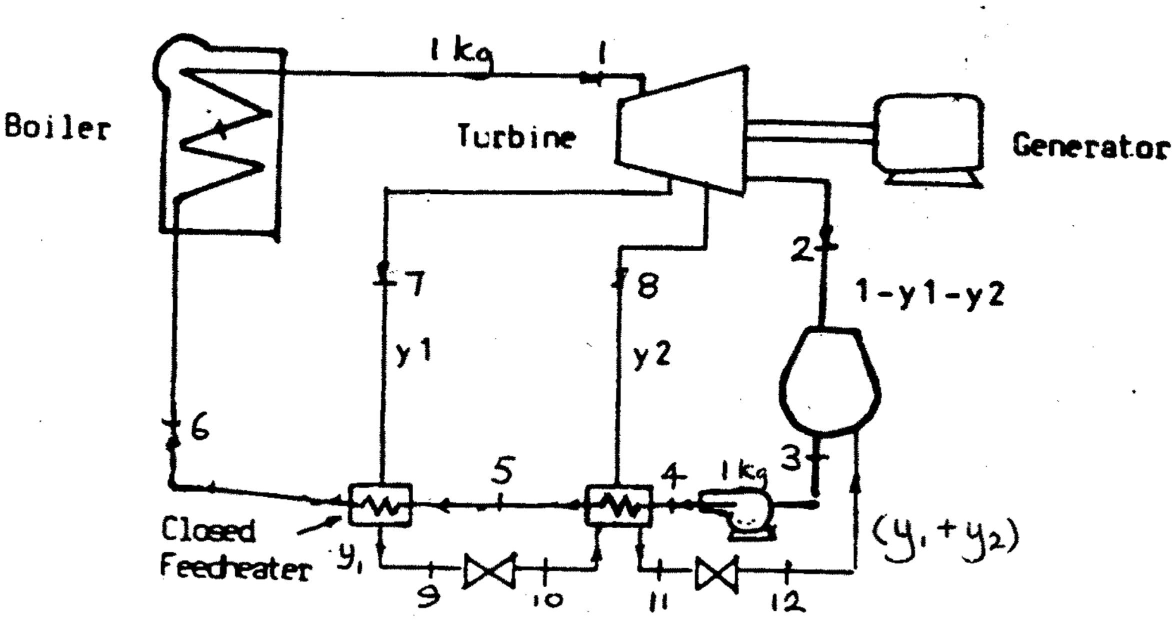
$$\sum \text{Enthalpy in} = \sum \text{Enthalpy out}$$
$$\sum m_i h_i = \sum m_o h_o$$
$$(1 - y_1)h_6 + y_1 h_9 = (1)h_8$$

which gives an equation that can be solved for y_1. Similarly, for heater 2,

$$\sum \text{Enthalpy in} = \sum \text{Enthalpy out}$$
$$\sum m_i h_i = \sum m_o h_o$$
$$(1 - y_1 - y_2)h_4 + y_2 h_{10} = (1 - y_1)h_5$$

which allows for y_2 to be solved, after knowing y_1.

It should be noted that these equations were only about open feed heaters. Closed feed heaters may be used instead, which means the resulting equation used to solve for y_1 and y_2 would be different.



For closed feedheaters, they act as heat exchangers, keeping the two inlets separate from each other. Although the two fluids do not mix with each other, they still do exchange heat and thus energy with each other. Thus, say for the left most feedheater, the equation is

$$\sum \text{Enthalpy in} = \sum \text{Enthalpy out}$$
$$\sum m_i h_i = \sum m_o h_o$$
$$y_1 h_7 + (1)h_5 = y_1 h_9 + (1)h_6$$

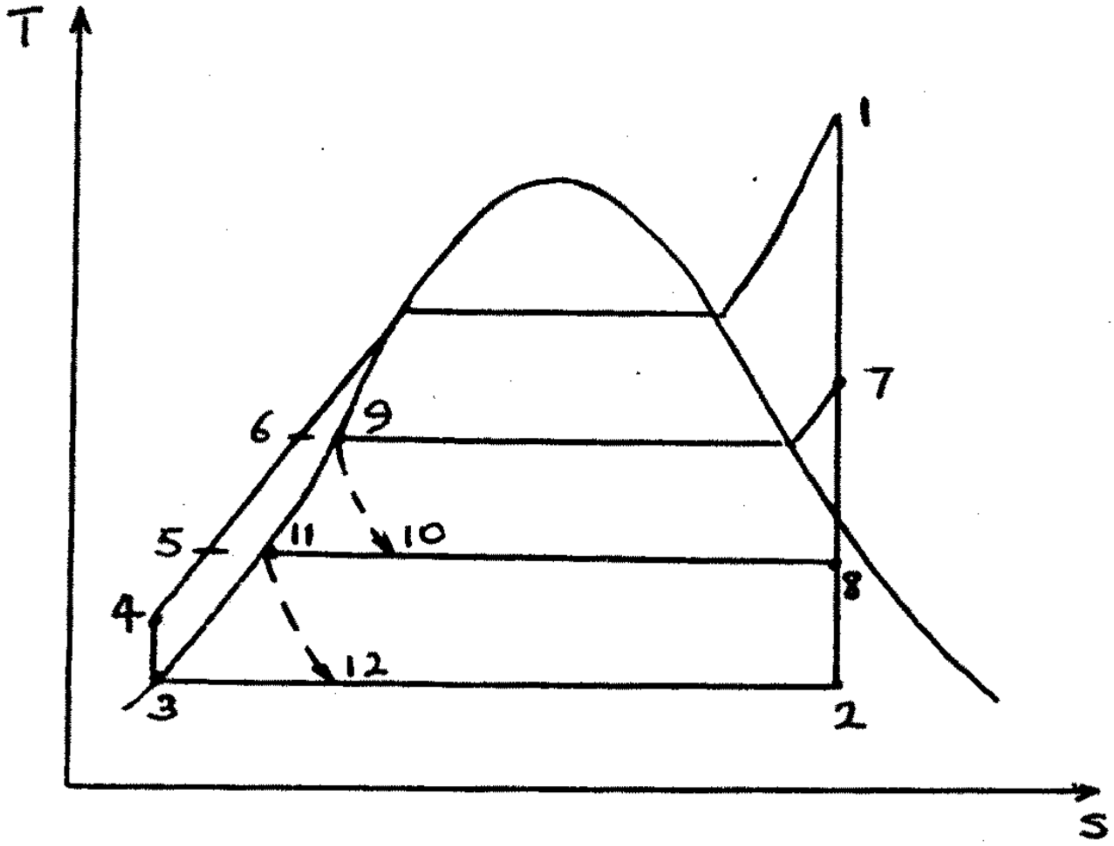
and the second feedheater gives

$$\sum \text{Enthalpy in} = \sum \text{Enthalpy out}$$
$$\sum m_i h_i = \sum m_o h_o$$
$$y_2 h_8 + (1)h_4 + h_{10} y_1 = (1)h_5 + (y_1 + y_2)h_{11}$$

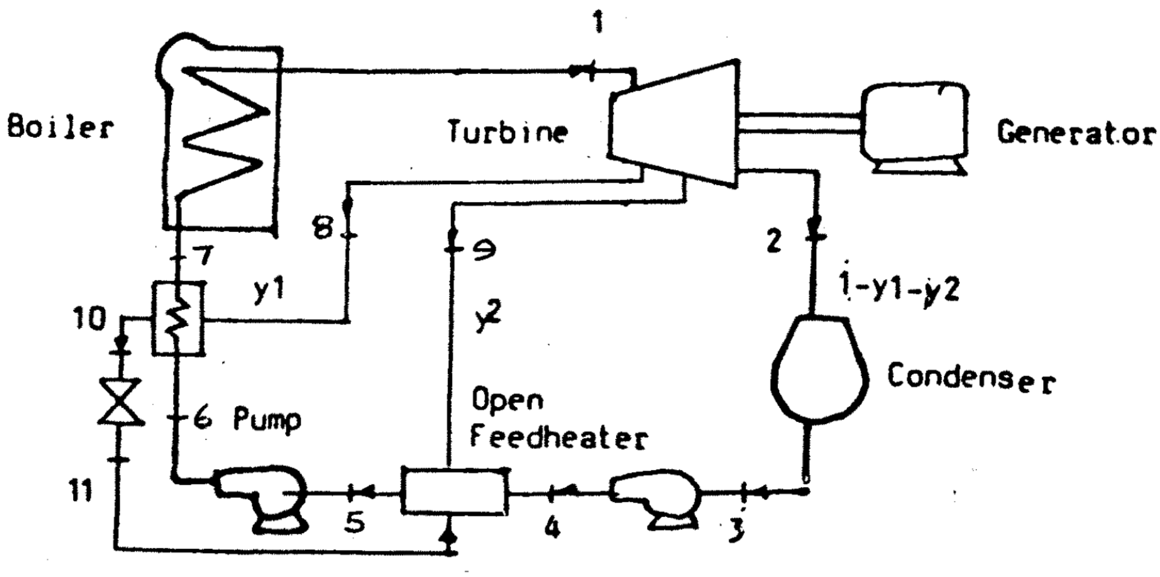
It should be noted that the feedheater outlet assumption of always outputting saturated liquid still holds true, thus

$$h_6 = h_9 \Rightarrow h_9 = h_{10}$$
$$h_5 = h_{11} \Rightarrow h_{11} = h_{12}$$

The resulting T-S diagram is also shown below.



In some cases, both open and closed feedheaters can be used, like so.



This means that the enthalpies and the bleed off proportions have to be found based on the setup, instead of following direct formulas.

After knowing the bleed off proportions and the enthalpies, the work produced by the turbine and the heat produced. For the typical regenerative cycle with open feedheaters, the heat produced is

$$Q = h_1 - h_8$$

while the work done is

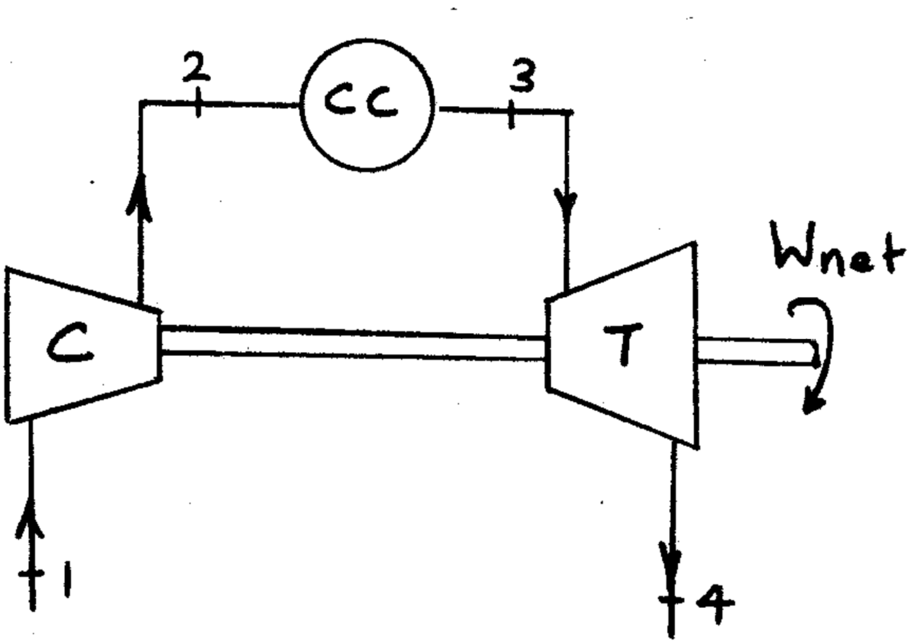
$$\begin{gathered} W = \sum \Delta h_j m_j \\ = (h_1 - h_9) + (1 - y_1)(h_9 - h_{10}) + (1 - y_1 - y_2)(h_{10} - h_2) \end{gathered}$$

Between open and closed feedheaters, there are some advantages and disadvantages of each feedheater. Because open feedheaters mix the steam together, it helps to act as a deaertion tank, allowing dissolved gases to be driven out of the fluid. Open feedheaters also allow for better heat transfer between the fluids as they are directly in contact with each other, instead of having a thin wall between them for closed feedheaters.

However, closed feedheaters have different advantages, in that they do not require a pump after the outlet to raise the pressure for the next heater/boiler, while open feedheaters do require it. Additionally, closed feedheaters are able to operate at higher pressures that open heaters, allowing for a higher enthalpy and work produced at the turbine.

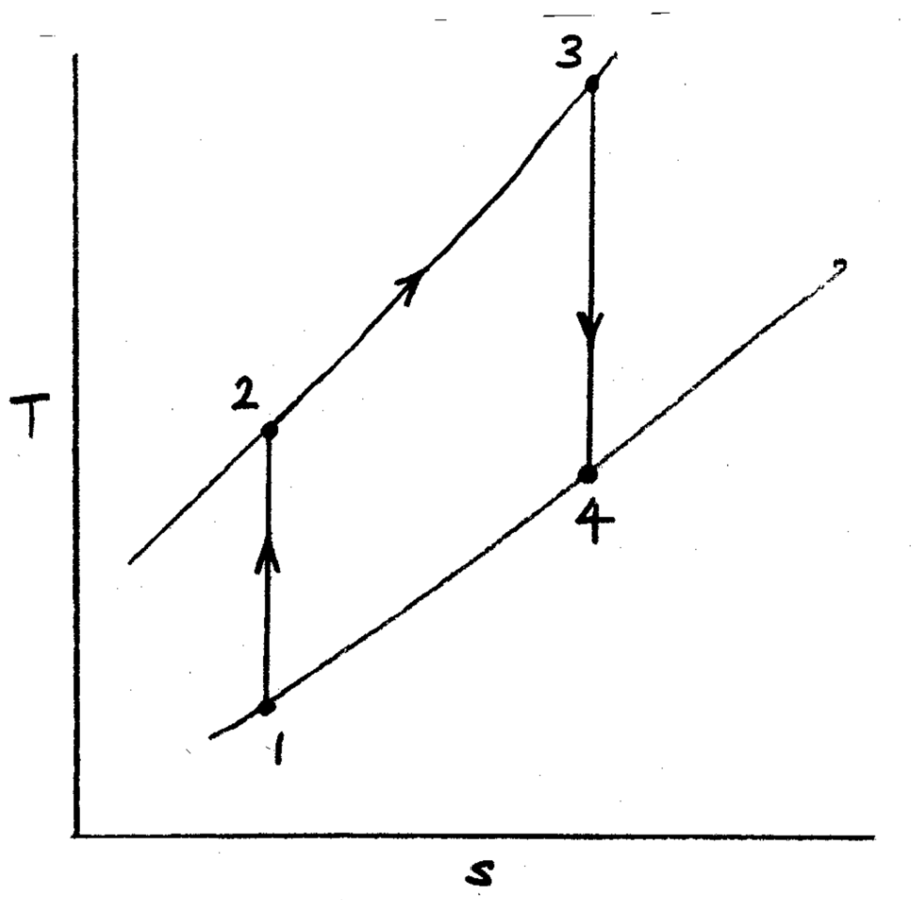
Gas Turbines

Recall from Thermofluids II that work can be done by combusting gas, allowing gas engines to produce work to power automobiles. In the case of power plants, a similar working principle is used to generate energy, by using a method similar to steam power plants, which is allowing gas to expand in a turbine. A typical gas turbine system is shown below.



Typically, gas turbines accept air at atmospheric pressure, before being isentropically compressed in a compressor, reaching a higher pressure which is then heated in the combustion chamber to a higher temperature, while still keeping the same pressure. Because of the higher pressure and temperature of the gas, the gas has a higher enthalpy and is able to release that energy as it expands through the turbine, before reaching the outlet where it is released back into the atmosphere. As the turbine allows the gas to expand, the turbine turns, which turns a shaft connected to a generator, as well as the compressor. This means that some work produced by the turbine is consumed by the compressor.

On the T-S diagram, the process looks like this



Note that this is not a closed process.

Generally, to find the efficiency of the gas turbine, the pressure and temperature of each point needs to be found. For the turbines, since we assume them to be isentropic, the pressure or temperature can be found from

$$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}}$$

Typically, the isentropic index for air is usually defined as 1.4. However, some extra considerations need to be made for gas turbines. For majority of the gas turbine, the gas passing through is typical air. However, after the air passes through the combustion chamber, it becomes contaminated with fuel from the combustion chamber. Hence, the gas from stage 2 to 3, and stage 3 to 4 is contaminated and will have a different isentropic index.

Typically, for the uncontaminated air, we use the symbol γ_a for air, while for contaminated air, we use γ_g for gas. These values are given as

$$\gamma_a = 1.4$$
$$\gamma_g = 1.333$$

Once the temperature of all the points are known, the heat produced by the combustion chamber, the work consumed by the compressor, and the work produced by the turbine can be found. To do this, we also need the specific heat capacity at constant pressure of air and gas, given as

$$C_{pa} = 1.005 kJ/kgK$$
$$C_{pg} = 1.15 kJ/kgK$$

From here, the values mentioned earlier can be found like so

$$Q = C_{pg}(T_3 - T_2) \text{ (heat gained)}$$
$$W_t = C_{pg}(T_3 - T_4) \text{ (work done)}$$
$$W_c = C_{pa}(T_2 - T_1) \text{ (work consumed)}$$

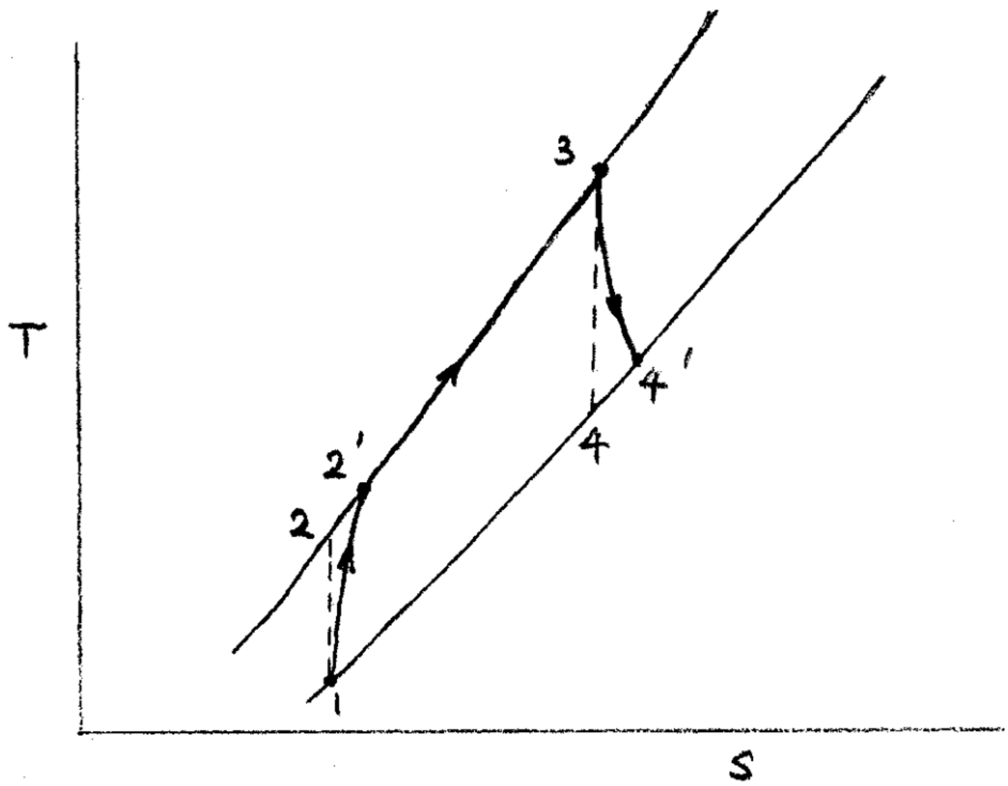
Then, the net work output is $W_t - W_c$. Using this, the thermal efficiency is

$$\eta = \frac{W_t - W_c}{Q}$$

and another metric, the work ratio, is the amount of net work produced against the turbine work, which is found as

$$W.R. = \frac{W_t - W_c}{W_t}$$

Now, this process had assumed the turbines and compressors are perfectly isentropic. However, turbines and compressors always have some isentropic efficiency to them. On the T-S diagram, since we know that all processes have $\Delta S > 0$, processes 1 to 2 and 3 to 4 will veer to the right slightly.



Hence, the actual temperature like T_2' or T_4' will vary slightly. To find this from the isentropic efficiency, we consider the actual work against the ideal work. For turbines, it is clear that ideal turbines would produce more work than actual turbines. Hence,

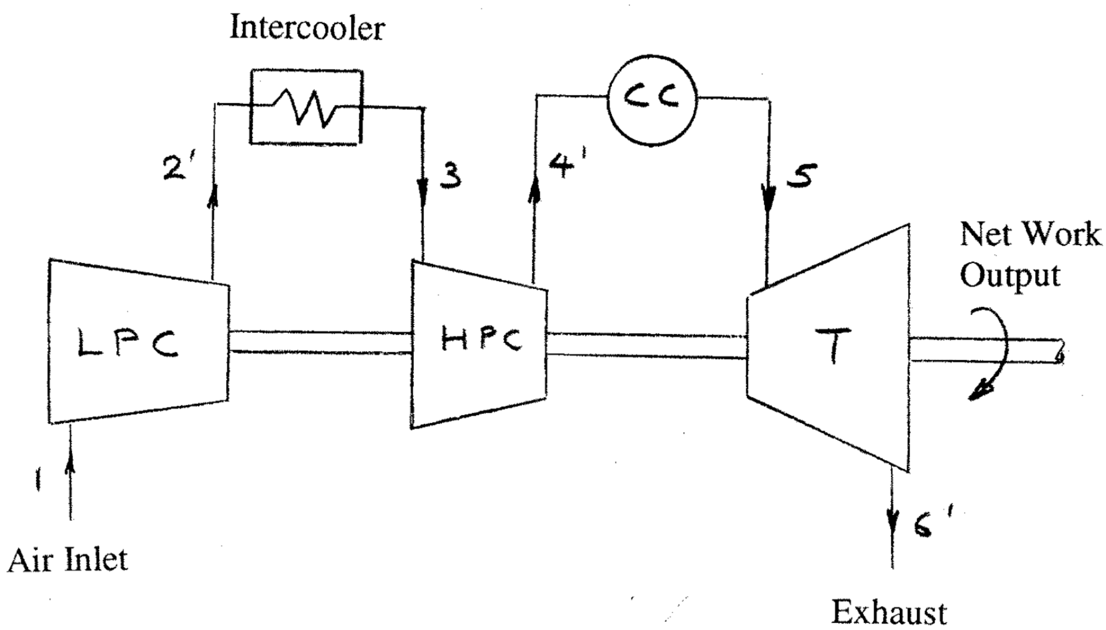
$$\eta_t = \frac{\text{actual work}}{\text{ideal work}}$$
$$= \frac{\cancel{C_{pg}}(T_3 - T_4')}{\cancel{C_{pg}}(T_3 - T_4)}$$
$$\therefore \eta_t = \frac{T_3 - T_4'}{T_3 - T_4} \Rightarrow T_4' = T_3 - \eta_t(T_3 - T_4)$$

As for compressors, ideal compressors actually consume less work than actuality, due to actual compressors needing more energy to overcome frictional losses. Hence, compressor efficiency is

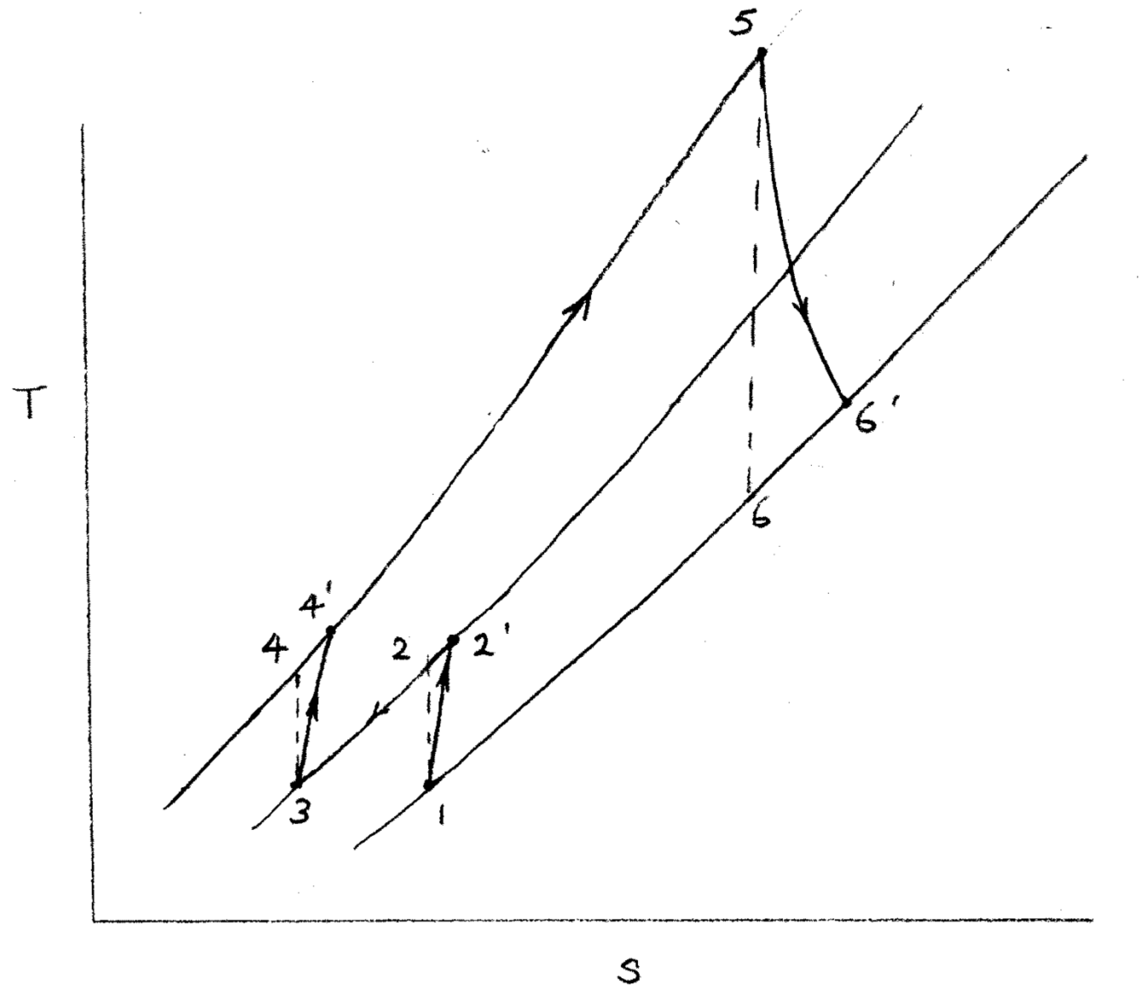
$$
\begin{aligned}
\eta_c &= \frac{\text{ideal work}}{\text{actual work}} \\
&= \frac{\cancel{C_{pa}}(T_2 - T_1)}{\cancel{C_{pa}}(T_2' - T_1)} \\
\therefore \eta_c &= \frac{T_2 - T_1}{T_2' - T_1} \Rightarrow T_2' = T_1 + \frac{T_2 - T_1}{\eta_c}
\end{aligned}
$$

In most cases however, gas turbines are not plainly setup with only a compressor, combustion chamber and turbine, as they tend to have lower work ratio and efficiency. Instead, different components are added, typically before the combustion chamber, to increase the efficiency of the gas turbine setup.

Intercooling works by splitting the compressor into two stages, one compressing from a lower pressure while one compressing at a higher pressure. Between the compressors, an intercooler is placed, which cools the gas from the lower pressure compressor back to the original atmospheric temperature.



If the intercooler is assumed to fully cool the gas back to atmospheric temperature, we say that the intercooling is complete, which means that $T_1 = T_3$.



For the compressors, if they both compress the gas by the same ratio, and we only knew the net compression pressure ratio, then the pressure ratio of each turbine is

$$\frac{P_2}{P_1} = \frac{P_4}{P_3} = \sqrt{P_{\text{overall}}} = \sqrt{\frac{P_4}{P_1}}$$

Because the pressure ratio is the same, $T_2 = T_4$.

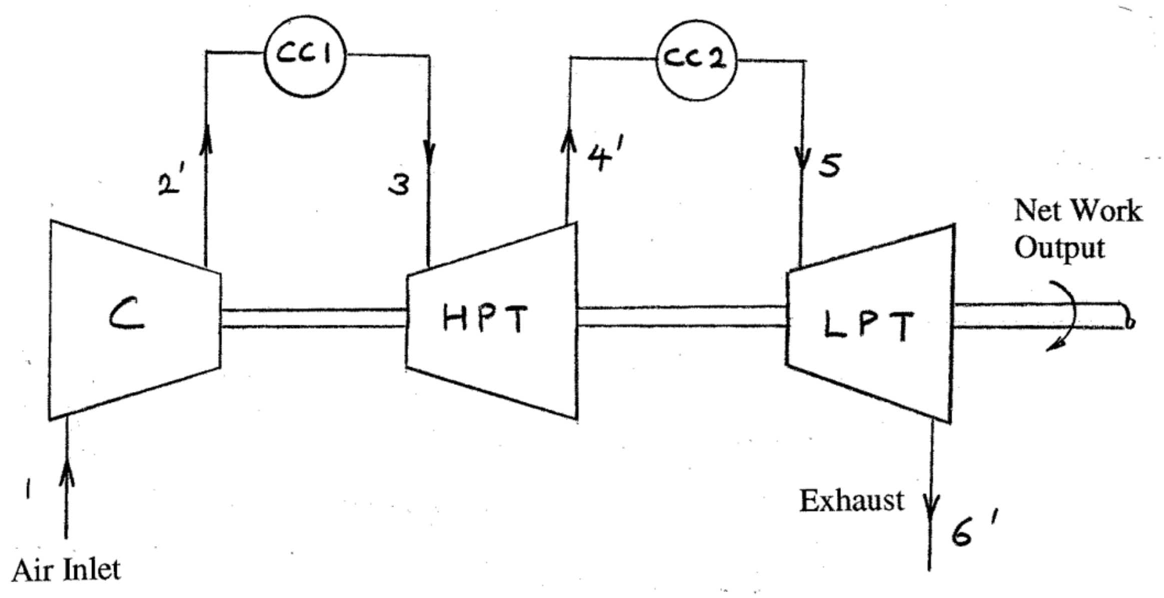
By implementing the intercooler, the net work consumed by both the compressors is lesser than one compressor, as the temperature difference is smaller, increasing the net work produced and hence the work ratio.

If both compressors also happen to have the same isentropic efficiency, then $T_2' = T_4'$. Thus, the work consumed by both the compressors is equal, meaning the net work consumed is

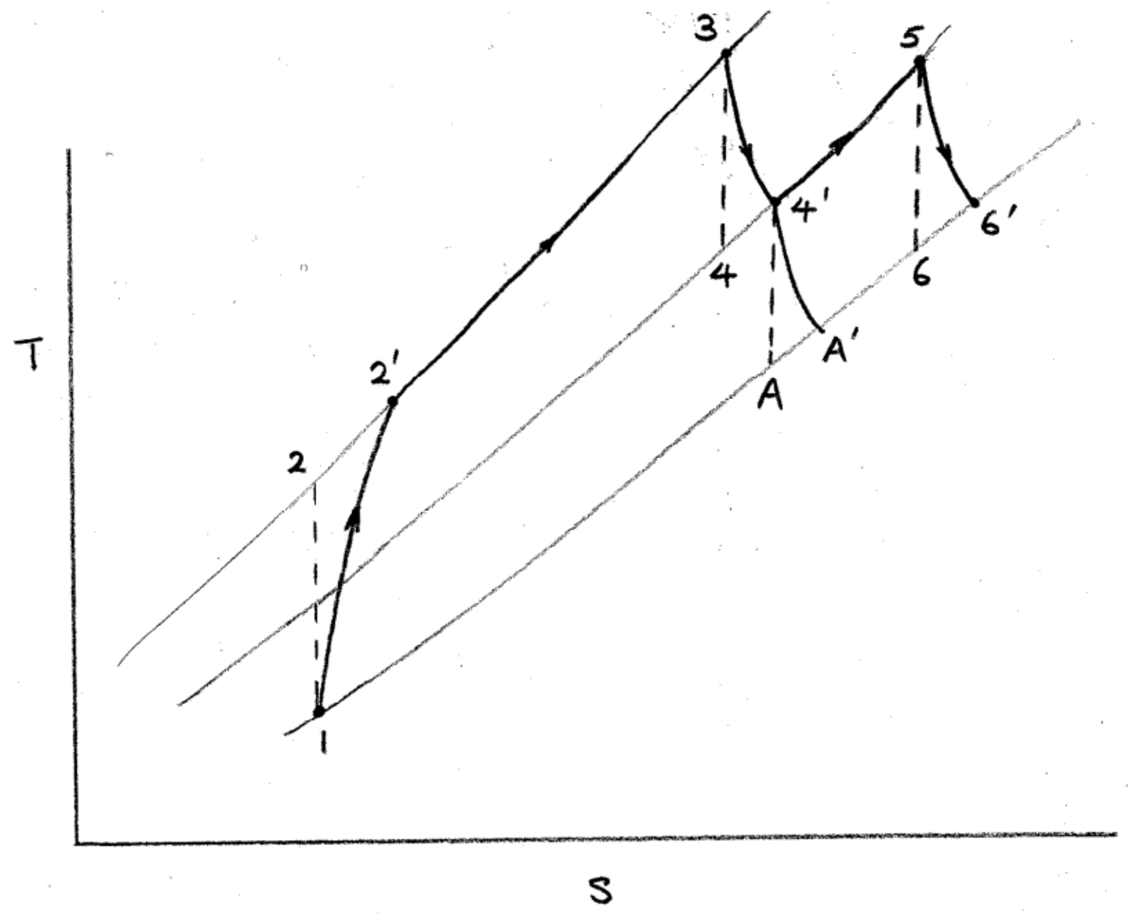
$$W_c = C_{pa}((T_2 - T_1) + (T_4 - T_3)) = 2C_{pa}(T_2 - T_1)$$

Although the work ratio will increase, the temperature of the gas going into the combustion chamber is lower than using a single compressor. Thus, more heat is supplied, and the efficiency may either increase or decrease.

A similar setup is the reheat setup, where the turbine is split into two instead. Then, a secondary combustion chamber is placed between the two turbines, which means the gas is heated twice through the whole process.

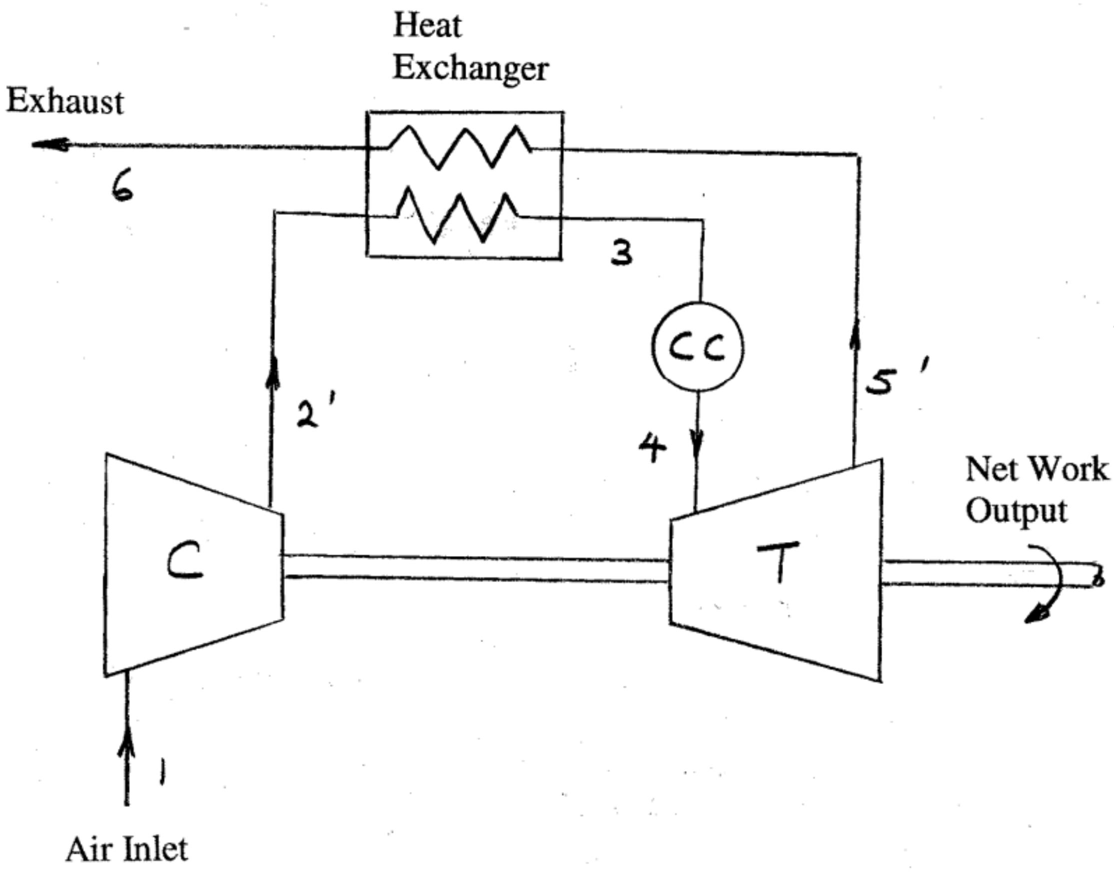


Since the outlet temperature of combustion chambers are given, the temperature and pressures of the system are not too difficult to find.



The effect of having two combustion chambers and turbines is that the isentropic efficiency of the turbines has less “effect” on the work done, as the distance 4′ and 6′ can veer will be lesser, allowing for more work to be done, increasing the work ratio. However, having two combustion chambers means that more heat will be produced, which can decrease the thermal efficiency.

To have a guaranteed increase in thermal efficiency, one method is by using a heat exchanger before the combustion chamber.



By adding a heat exchanger, the warmer turbine outlet gas is able to heat up the compressor outlet gas slightly, reducing the heat needed from the combustion chamber, thus causing the efficiency to be higher. Note that both the heat exchanger and the combustion chamber still heat the gas isobarically.

To determine how much the heat exchanger has heated up the compressor outlet gas, we use the thermal ratio. Ideally, all the heat from the warmer gas in 5 to 6 heats the gas from 2 to 3. Hence, if the heat exchanger was infinitely long for example, then the outlet temperature of the heat exchanger (T_3) would be the same as T_5. This means that the maximum possible rise in the temperature for the cold side would be $T_5 - T_2'$. However, the actual increase is only $T_3 - T_2'$. Hence, this forms the thermal ratio, given as

$$T.R. = \frac{\text{actual temp rise}}{\text{max temp rise}} = \frac{T_3 - T_2'}{T_5 - T_2'}$$